

HT Group	Name
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CATHOLIC JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS

H1 BIOLOGY

8875/02

Paper 2 Core

27 August 2008 Wednesday

2 hours

Additional materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer **one** question.

Write your answers on the separate answer paper.

All working for numerical answers must be shown.

At the end of the examination,

1. fasten all your work securely together;
2. enter the number of the Section B question you have answered in the grid opposite.

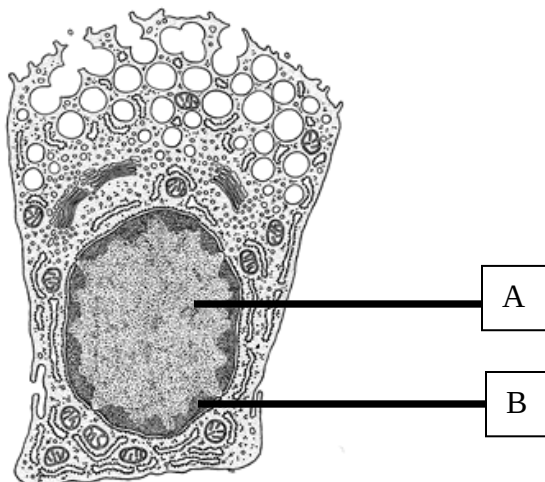
The number of marks is given in brackets [] at the end of each question or part question.

Section A	
1 [10]	
2 [13]	
3 [12]	
4 [15]	
5 [17]	
6 [13]	
Section B	
Total	

This document consists of **11** printed pages.

Section A – Answer ALL questions

- 1 The diagram below shows a secretory cell from the liver.



- (a) State the names of part A and B [2]

A: _____

B: _____

- (b) Explain how the staining patterns in the nucleus are related to the cell's activity. [2]

Some cells similar to that shown in the diagram were grown in a culture. Radioactive amino acids were added to the solution in which they were being grown. The radioactivity acts as a label on the amino acid so that it can be detected wherever it is. This radioactive label allows amino acids to be followed through the cell.

At various times, samples of the cells were taken and the amount of radioactivity in different organelles was measured. The results are shown in the table.

Time after radioactive amino acids were added to the solution/ minutes	Amount of radioactivity present/arbitrary units		
	Golgi apparatus	Rough endoplasmic reticulum	Vesicles
1	21	120	6
20	42	68	6
40	86	39	8
60	76	28	15
90	50	27	28
120	38	26	56

(c) What is the fate of amino acids that reach the endoplasmic reticulum? [2]

(d) Account for the difference in radioactivity observed in the organelles after 60 minutes. [4]

[Total: 10]

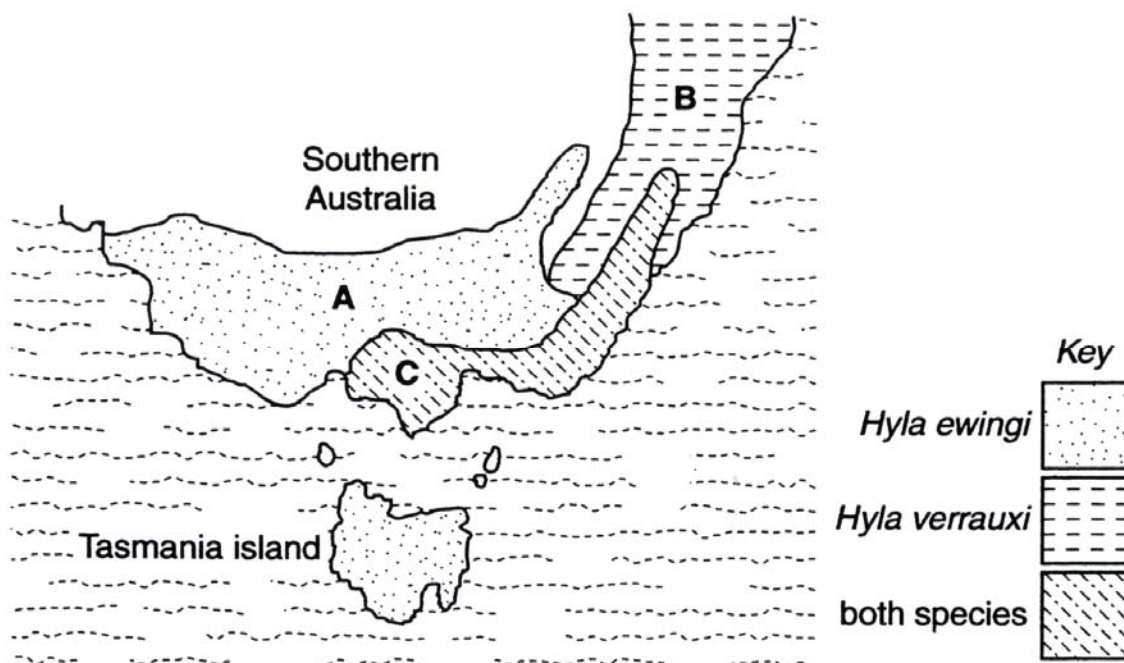
2 Two closely related species of frog, *Hyla ewingi* and *Hyla verrauxi* live in South Australia.

(a) Explain what is meant by the terms;

(i) *Niche* [2]

(ii) *Species* [3]

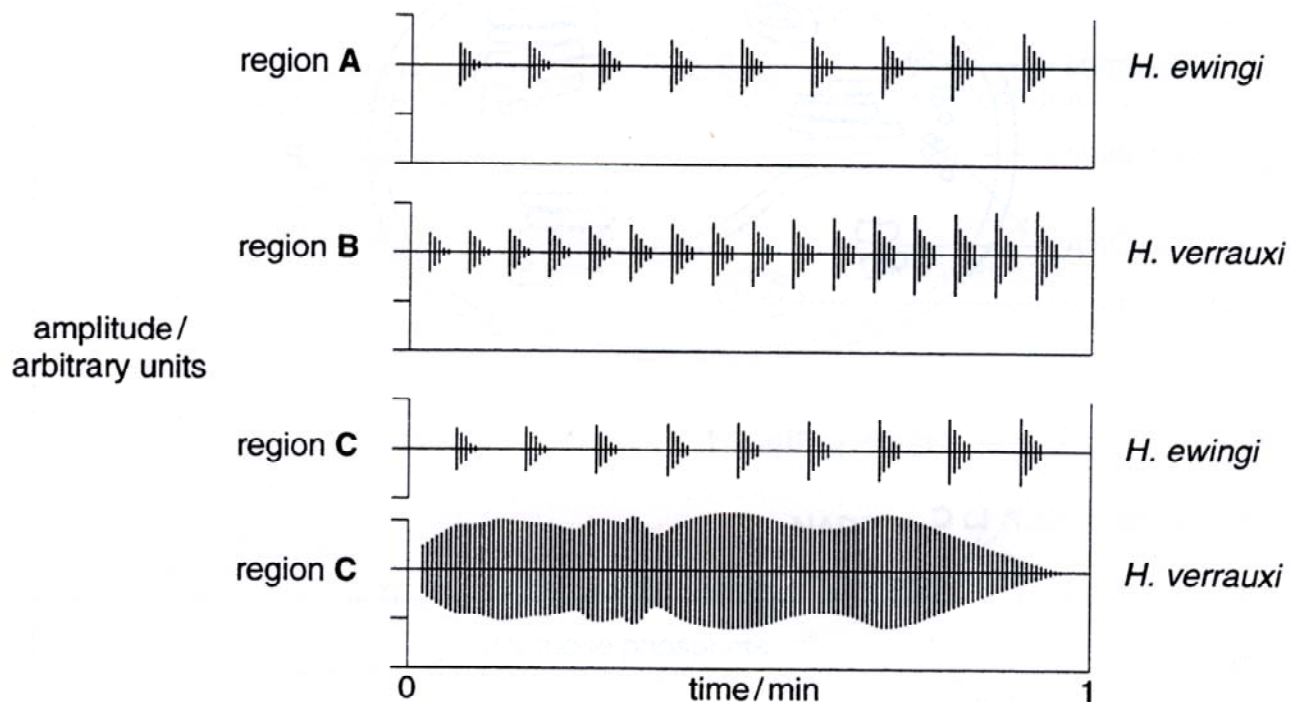
The figure below shows the distribution of the tree frogs in Southern Australia.



Hyla ewingi and *Hyla verrauxi* are two closely related species of tree frogs from southern Australia. Research from breeding studies and DNA sequence data has shown that they have weak genetic incompatibility.

Male frogs attract females of the same species for mating by their pulsing call. The pulse rate of the male calls of the two species is almost identical. However, when both species coexist within the same region, the calls of *H. ewingi* are quite different than those of *H. verrauxi*.

The following data shows the calls of *H. ewingi* and *H. verrauxi* from three regions, A, B and C. The male advertisement calls were recorded and played back electronically to produce the pattern shown



Some female frogs of the species *H. verreauxi* were transferred from region B to region C. Observations shown that they were not attracted by the calls of the males in region C.

(b) Suggest why this is so. [3]

(c) Explain how the process of natural selection could have led to the different male advertisement call in *H. verreauxi* in region C. [5]

[Total: 13]

3 Fig. 3.1 and 3.2 are diagrams showing transcription and translation.

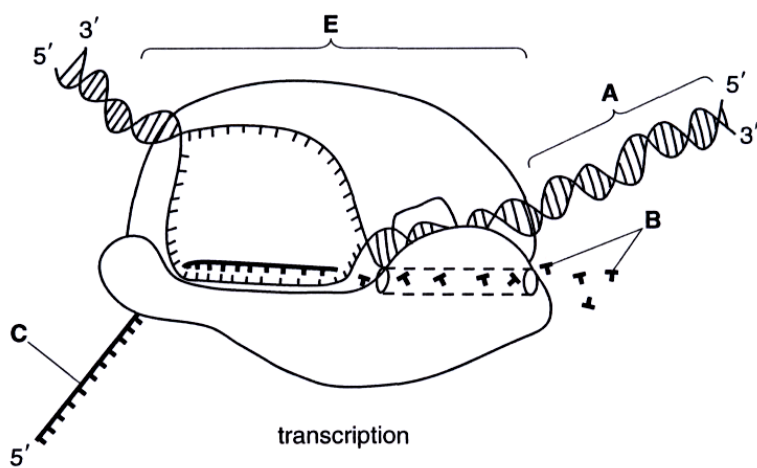


Fig. 3.1

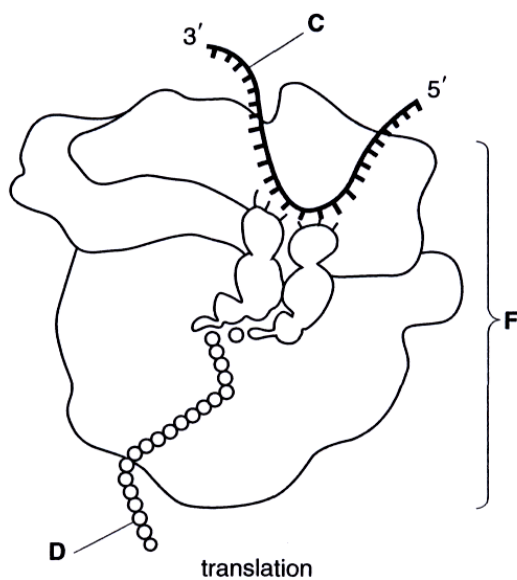


Fig. 3.2

(a) Identify structures A to D and describe their function. [4]

A: _____

B: _____

C: _____

D: _____

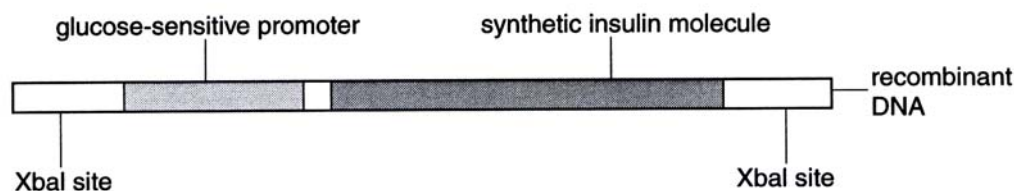
(b) State where in the cell does transcription and translation occurs. [2]

- (c) List four additional ways in which transcription differs from translation. [4]

- (d) Briefly describe how structure F differs from structure E. [2]

[Total: 12]

- 4 Using genetic engineered bacteria, a synthetic insulin molecule has been produced which has over 20% of the activity of naturally produced insulin. The gene for the synthetic molecule has been used in an experimental gene therapy for insulin – dependent diabetes. DNA coding for the synthetic molecule was combined with a promoter region (binding site for RNA polymerase during transcription) taken from a gene for an enzyme found only in liver cells. The promoter is sensitive to blood glucose concentration and the synthetic insulin gene will become transcribed when blood glucose concentration rises above normal concentration.



- (a) Describe the role of restriction enzymes, like XbaI in genetic engineering. [4]

- (b) Describe how a bacterium that takes up a genetically modified plasmid can be identified. [4]

Insulin is also synthesized in large scale fermenters from genetically modified bacteria.

- (c) Explain advantages to a patient of using insulin produced by genetically modified bacteria. [3]

- (d) Insulin is a peptide hormone consisting of 51 amino acids. Describe the size of the coding region of the synthetic molecule in the diagram above. Show your working. [2]

- (e) Suggest an alternative organism that can be used to produce human insulin by genetic transformation. Give a reason for your answer. [2]

[Total: 15]

- 5 In 1882, T. W. Englemann investigated the effect of different wavelengths of light on photosynthesis. He placed a filamentous green alga into a test-tube along with a suspension of motile bacteria which move to regions of high oxygen concentration. He allowed the bacteria to use up the available oxygen and then illuminated the alga with light that had been passed through a prism to form a spectrum. After a short time, he observed the results shown in Fig. 5.1. Bacteria, which are indicated by the tiny rectangles, were distributed evenly throughout the test-tube at the beginning of the experiment.

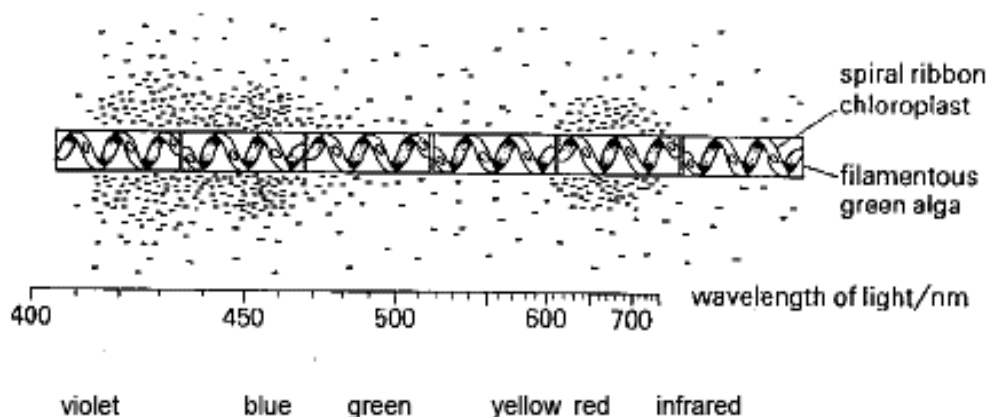


Fig. 5.1

- (a)** Using the information provided above, describe how photosynthesis varies with the wavelength of light. [2]

- (b)** With reference to light dependent reactions, explain the results observed in Fig. 5.1. [4]

- (c)** State and explain the difference in observations if water supply to the algae was limited. [3]

Photosynthesis in the green alga occurs in two stages, at the thylakoid membrane (light dependent) and the stroma (light independent reactions.)

- (d)** Describe the function of the thylakoid membrane in photophosphorylation. [3]

- (e) Describe the differences in photophosphorylation and oxidative phosphorylation in chloroplast and mitochondria. [5]

[Total: 17]

- 6 In the garden pea, *Pisum sativum* two pairs of alleles determine the seed characters green and yellow and round and wrinkled. Table 6.1 shows the results which were obtained from four separate crosses.

Cross	Parent	Progeny			
		Yellow round	Yellow wrinkled	Green round	Green wrinkled
1	Yellow round x green wrinkled	138	143	137	142
2	Yellow round x yellow round	176	0	60	0
3	Yellow round x yellow wrinkled	223	247	75	86
4	Green round x yellow wrinkled	371	0	0	0

- (a) State which alleles are dominant and explain your answer. [2]

- (b) Using suitable symbols, write down the genotypes of the parents of each cross in the table below. Show your working. [8]

Symbols used:

Parents of Cross 1:

Parents of cross 2:

Parents of cross 3:

Parents of cross 4:

Another experiment was carried out where randomly selected round, yellow plants and green wrinkled plants were self fertilized, and it was found out that only green, wrinkled plants always produced pure-breeding offspring.

- (c) Explain why wrinkled, green plants always produce pure breeding offspring while round yellow plants do not. [3]

[Total: 13]

Section B

Answer ONE question ONLY.

Write your answer on the separate answer paper provided.

Your answers must be in complete prose, where appropriate.

Your answers must be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be set out in sections (a), (b) etc, as indicated in the question.

- 7 (a) Describe how prokaryotic genome structure differs from eukaryotes, and explain its significance in gene cloning. [10]
- (b) Using named examples, explain the properties of stem cells. [6]
- (c) Discuss the ethical issues involved with the production of genetically modified organisms. [4]
- 8 (a) Outline the main features of the Krebs's Cycle. [8]
- (b) Describe the role of NAD in aerobic respiration. [6]
- (c) Explain the small yield of ATP from anaerobic respiration. [6]

-END OF PAPER-